Department of Computer Science
Faculty of Computing

Project Task 2

(*Initial Design - Preliminary Analysis*)

Programme	Bachelor of Computer Science (*Data Engineering*)
Subject Code	SECR1213
Subject Name	Network Communications
Session-Sem	2024/2025-1
Prepared by	1) ANIS SAFIYYA BIN JANAI 2) NURUL ASYIKIN BINTI KHAIRUL ANUAR 3) DAMIYA AINA BINTI BASIR ABD SHAMMAD 4) AIN NURNABILA BINTI MOHD AZHAR
Section	02
Group	CircuitCrew (CC)
Lecturer	DR. ISMAIL FAUZI BIN ISNIN
Date	9 / 11 / 2024

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Question

1. What is needed for each lab?

Network switches should be installed in the Embedded Lab so that workstations can connect to the local network. For network segmentation and internet access, a router will be necessary. In order to connect embedded devices and systems within the lab, Ethernet connections and cabling are required. Students will have flexible connectivity options and wireless device compatibility thanks to a Wi-Fi access point. To enable students to work with server-side applications, a small server rack might be included as an optional feature. In order for students to have practical experience configuring and troubleshooting network hardware, the CISCO Lab will require CISCO routers and switches. For convenience of reconfiguring network connections, patch panels are advised. There should be enough Ethernet cabling accessible for hands-on practice with wired network configurations. Students will also benefit from using a Wi-Fi access point to experience configuring and maintaining wireless networks.[1]

For group projects and presentations, the hybrid classroom should feature a projector or display screens. Students' devices will be able to access the internet through a Wi-Fi access point, and Ethernet ports should be accessible for dependable, fast connections, particularly during demonstrations. Furthermore, video conferencing technology will make distant or hybrid learning possible, creating a dynamic and adaptable learning environment. For student assignments, network testing, and simulations, the General Lab requires PCs with specific networking software. Research, online resource access, and internet-based application testing all require a fast internet connection. Installing Ethernet ports and Wi-Fi access points will give students a variety of connectivity choices and give them freedom when it comes to network access.[1]

2. What security measures are required to protect against potential network breaches like Internet Worms, denial-of-service attacks, and e-business application attacks?

A firewall and network segmentation will separate this area from the rest of the network in the CISCO Lab, preventing Internet worms from spreading. Strict physical access control will stop unwanted configuration modifications, and an intrusion detection system (IDPS) will keep an eye out for odd activity. Protecting lab equipment from DDoS attacks will enable students to practice safely. In order to prevent unwanted access, the Embedded Lab should have access control lists, a local firewall, and network segmentation for IoT devices. Data between devices will be protected by SSL/TLS encryption, and any anomalous activity will be identified by centralized logging. An intrusion detection system and secure Wi-Fi with WPA3 are essential for the General Lab in order to monitor traffic and stop unwanted access. Antivirus software and role-based access control (RBAC) will guard against dangerous software and insider threats.Secure Wi-Fi, RBAC, and device endpoint security will restrict access to approved apps and identify malware in the hybrid classroom. Lastly, Network Access Control (NAC) will guarantee that only compliant devices connect, and a guest Wi-Fi network in the student waiting area and common areas will separate visitor devices from the main network. The network will be shielded from denial-of-service assaults, Internet worms, and application vulnerabilities in several domains by these focused security measures.[5]

3. What are the expected data speeds and bandwidth requirements for each area, particularly the labs and the video conferencing room?

The CISCO Lab requires 1 Gbps speeds and 500 Mbps to 1 Gbps bandwidth for heavy network tasks. In the Embedded Lab, 100-500 Mbps bandwidth is adequate, with some 1 Gbps connections for higher demands. The General Lab needs 100-500 Mbps bandwidth for research and simulations, while the Video Conferencing Room should have 200-300 Mbps to support HD and 4K video smoothly. The Hybrid Classroom needs 100-200 Mbps with both Ethernet and Wi-Fi to support flexible learning. These configurations ensure reliable performance tailored to each area's activities.[3]

4. How many additional users (students, faculty, or devices) should the network support in the future as FC grows?

The network should accommodate 15% more than it does now in order to accommodate more students, professors, and devices in the coming years. The network should be able to accommodate more than 2000 users if it currently supports 1080 people. Particularly in crowded places like labs and common areas, each user may connect two to three devices. The network will be able to manage this expansion more easily with scalable infrastructure.[9]

5. What are the expected software and hardware requirements for each lab, especially for the Cisco Network Lab and the Embedded Lab?

The Embedded Lab requires microcontrollers, development boards, moderate-power computers, and software for embedded programming; the Cisco Network Lab requires Cisco routers, switches, and workstations with network simulation software such as Cisco Packet Tracer; and the General Labs require standard computers with general-purpose software for productivity and simulations.[1]

6. What type of wireless network coverage is expected in areas such as the student lounge, and how much capacity will it need?

The best choice for a student lounge is a Wireless LAN (WLAN) since it provides dependable, fast connectivity in a small to medium-sized indoor space. In settings like a student lounge, where numerous users must use the network concurrently, WLANs' ability to support multiple devices at once is crucial. WLANs are perfect for spaces that don't need the extensive infrastructure of wider area networks because their coverage range usually covers an entire building or room. Furthermore, compared to other wireless solutions like MANs or WANs, which are intended for greater geographic areas, WLANs are less expensive to establish and maintain.[2]

7. Are there any limitations on the types of devices that can be used within the network due to compatibility or budget concerns?

Indeed, there are restrictions. Because older devices may not be compatible with modern Wi-Fi standards and security mechanisms, their performance may be slower. By restricting access points and management tools, financial limitations can lower network performance and coverage. High-demand activities may also be restricted by bandwidth constraints, and devices that do not support PoE may require more power.[8]

8. How scalable does the system need to be for future expansion, and are there specific future upgrades planned?

For future expansion in the Faculty of Computing(FC) we are considering the advancement of a network which is Wi-Fi 6E or Wi-Fi7. This advancement is crucial as the faculty's students rely on high-speed, reliable internet for their studies that require data-intensive applications such as cloud computing and artificial intelligence. Moreover, the selected Wi-Fi is predicted to be ten times faster than what 5G requires that will support more devices, making it suitable for high-density areas like labs and student lounge[6].

Furthermore, expanding the user base and providing access to more sophisticated features that future projects may need will be made possible by updating software solutions and licensing for academic and research applications. A more dynamic and cooperative learning environment can be produced by improving classroom technology, such as with smart classroom solutions that include interactive whiteboards, AR/VR integration, and AV systems[7].

9. Are there any environmental or infrastructure challenges (e.g., power, space constraints) that might impact network design?

Since the focus of this project was the building of the Faculty of Computing, there might be a space constraint. This is because there is limited physical space for the equipment such as servers, switches and cabling.It might be difficult to arrange equipment in a small area to provide effective cooling and appropriate power distribution without interfering with operations or limiting future growth potential[4].

Other than that, the Faculty of Computing's network architecture may be impacted by a number of additional environmental and infrastructure issues in addition to space limitations. Because network devices like servers, routers, and switches need steady power sources to avoid downtime and data loss, power supply dependability is essential. Regular power outages can interfere with system access and research activities, which affects the network's overall performance.

Furthermore, organizing and managing cables might be difficult. Inadequately managed wires may prevent ventilation, raise the possibility of fire dangers, and complicate maintenance in a small space. Effective cable organization can be achieved while maximizing safety and airflow by putting in place structured cabling systems, such as overhead or underfloor cable trays.

10. What is the projected timeline for implementing the network, and are there any time-sensitive milestones?

In the one to two month planning and design phase, equipment security, network design finalization, and space and power requirements are evaluated. This phase is essential for identifying potential environmental difficulties, such as space constraints and power supply limitations. Time-sensitive milestones during this time include confirming the availability of space and starting the equipment selection procedure to prevent delays in installation.

The physical infrastructure, including servers, switches, cabling, and power systems, must be set up during the installation and configuration phase, which lasts two to four months. During the testing and optimization phase, which lasts 1 to 2 months, the network is tested for performance, security protocol implementations, and system optimization. This stage should make sure that the network can manage heavy traffic, particularly in places with a lot of people, like labs.To guarantee the network's long-term dependability and functionality, constant maintenance and monitoring will be required. This phase will be implemented after the deployment phase.

Feasibility Assessment

The planned network infrastructure project at the Faculty of Computing has undergone a thorough study of its viability from an operational, economic, and technological standpoint. Through this evaluation, the network architecture is guaranteed to accommodate future expansion and technological developments while satisfying present administrative and academic needs. The feasibility assessment assesses the infrastructure's capacity to support growing user numbers, conform to emerging technologies, and sustain high functionality within financial restrictions, with particular attention to scalability, security, and efficient performance.

Technical Feasibility:

The suggested network infrastructure must satisfy both present academic requirements and future scalability for the project to be technically feasible. The design takes into account the various needs of labs and the hybrid classroom by integrating Ethernet and Wi-Fi networking and guaranteeing high-speed internet access. The network design is technically feasible for a growing institution since it accounts for future-proof technologies like Wi-Fi 6E or 7 and anticipates rising demand and sophisticated technological capabilities.

Economic Feasibility:

By emphasizing high-performing yet reasonably priced devices, the suggested infrastructure provides a balanced approach that satisfies both present and projected future demands without going over budget. The concept is economically feasible in promoting academic progress, even though cutting-edge technologies like improved wireless networks and security measures raise initial expenses. However, they guarantee long-term savings by lowering the need for periodic upgrades.

Operational Feasibility:

The network design's operational goal is to improve teacher and student usability by putting in place a simplified, manageable infrastructure. In line with institutional objectives, the addition of Wi-Fi for flexible access and safe lab configurations facilitates efficient academic operations as well as real-world learning scenarios. It is anticipated that the network will satisfy the Faculty of Computing's daily operational requirements by successfully addressing security and accessibility.

Security Considerations:

In order to stop unwanted access and lessen the risk of cyberattacks such malware, denial-of-service (DoS) assaults, and Internet worms, the suggested network design revolves around a strong, multi-layered security operation. Network segmentation and specialized firewalls will separate sensitive areas from other traffic in each lab and classroom, greatly lowering the possibility of lateral movement after a security compromise. SSL/TLS encryption for data transfer and WPA3 for Wi-Fi are examples of advanced security technologies that guarantee secure communications. Network access control (NAC) and role-based access control

(RBAC) are two examples of access control mechanisms that further limit user rights, protecting the network from internal and external threats.

Meeting Minutes #1

<table>
<tr><td colspan="2">Date/Time</td><td colspan="2">3 Nov 2024 / 9.00 pm</td></tr>
<tr><td colspan="2">Location</td><td colspan="2">Online (Google Meet)</td></tr>
<tr><td colspan="2">Agenda</td><td colspan="2">Task Distribution for Task 2</td></tr>
<tr><td colspan="2">Meeting Host</td><td colspan="2">Ain Nurnabila binti Mohd Azhar</td></tr>
<tr><td colspan="4" align="center">Attendance</td></tr>
<tr><td colspan="2">Name</td><td>Time</td><td>Reason For Absence</td></tr>
<tr><td colspan="2">Anis Safiyya</td><td>9.00 pm</td><td>-</td></tr>
<tr><td colspan="2">Nurul Asyikin</td><td>9.00 pm</td><td>-</td></tr>
<tr><td colspan="2">Damiya Aina</td><td>9.00 pm</td><td>-</td></tr>
<tr><td colspan="2">Ain Nurnabila</td><td>9.00 pm</td><td>-</td></tr>
<tr><td colspan="4" align="center">Minutes</td></tr>
<tr><td>No.</td><td>Item Discussed</td><td>Details</td><td>Person-In-Charge</td></tr>
<tr><td>1.</td><td>Find group name</td><td>All members discussed and compromise for the task distribution</td><td>All members</td></tr>
<tr><td>2.</td><td>Task distribution</td><td>- All of us study the task
- Confirm the number of the question needed for this task</td><td>All members</td></tr>
<tr><td>3.</td><td>Question listing</td><td>- Ain in charge of questions listing</td><td>Ain Nurnabila</td></tr>
<tr><td>3.</td><td>Methods for information gathering</td><td>- Damiya, Anis and Asyikin divide the questions listed by Ain</td><td>All members</td></tr>
<tr><td>4.</td><td>Meeting ended</td><td>- At 10.15 pm, Ain ended the meeting</td><td>Ain Nurnabila</td></tr>
</table>

		- Everyone contributed well for this meeting.	

Meeting Minutes #2

<table>
<tr><td colspan="2">Date/Time</td><td colspan="2">6 Nov 2024 / 1 pm</td></tr>
<tr><td colspan="2">Location</td><td colspan="2">N28, Faculty of Computing</td></tr>
<tr><td colspan="2">Agenda</td><td colspan="2">Progress Discussion</td></tr>
<tr><td colspan="2">Meeting Host</td><td colspan="2">Nurul Asyikin binti Khairul Anuar</td></tr>
<tr><td colspan="4" align="center">Attendance</td></tr>
<tr><td colspan="2">Name</td><td>Time</td><td>Reason For Absence</td></tr>
<tr><td colspan="2">Anis Safiyya</td><td>1.15 pm</td><td>-</td></tr>
<tr><td colspan="2">Nurul Asyikin</td><td>1.15 pm</td><td>-</td></tr>
<tr><td colspan="2">Damiya Aina</td><td>1.00 pm</td><td>-</td></tr>
<tr><td colspan="2">Ain Nurnabila</td><td>1.15 pm</td><td>-</td></tr>
<tr><td colspan="4" align="center">Minutes</td></tr>
<tr><td>No.</td><td>Item Discussed</td><td>Details</td><td>Person-In-Charge</td></tr>
<tr><td>1.</td><td>Sharing session on current progress</td><td>All of the members shared the current progress</td><td>All members</td></tr>
<tr><td>2.</td><td>Discuss on the answer for the questions</td><td>- All members share ideas and opinions for each question</td><td>All members</td></tr>
<tr><td>3.</td><td>Additional information</td><td>- Each member help each other to elaborate in detail for all questions</td><td>All members</td></tr>
<tr><td>4.</td><td>Meeting ended</td><td>At 3.00 pm, Damiya ended the meeting</td><td>Damiya Aina</td></tr>
</table>

REFERENCE

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